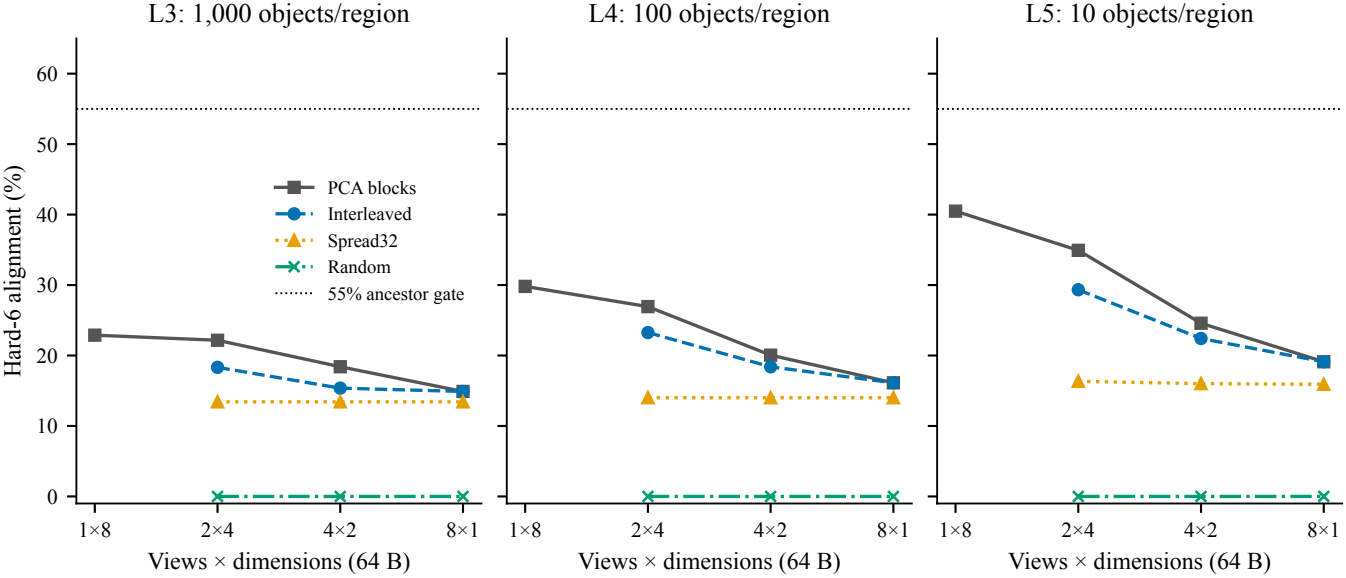
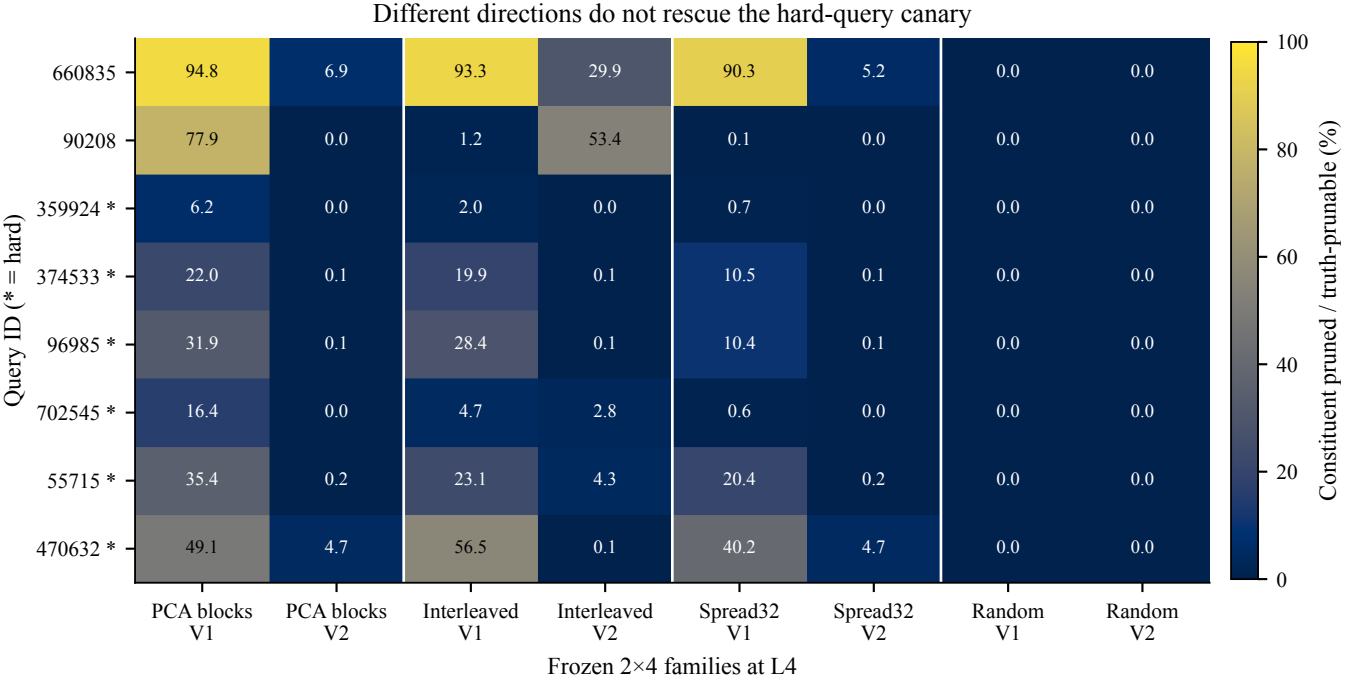
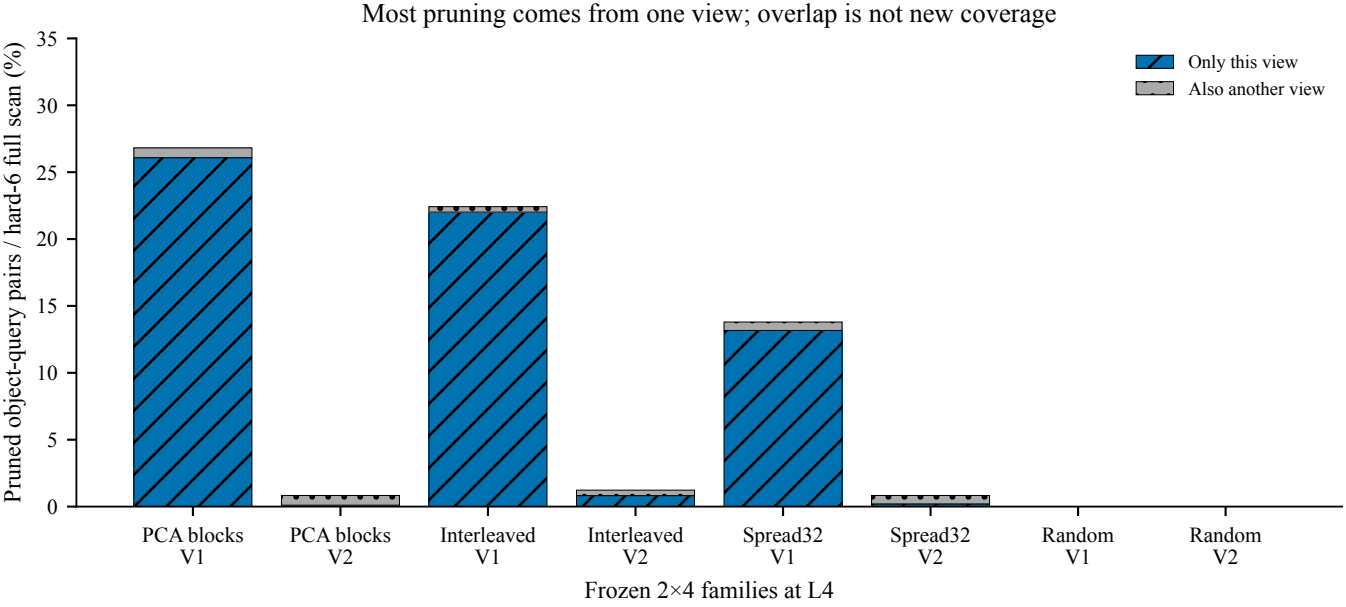
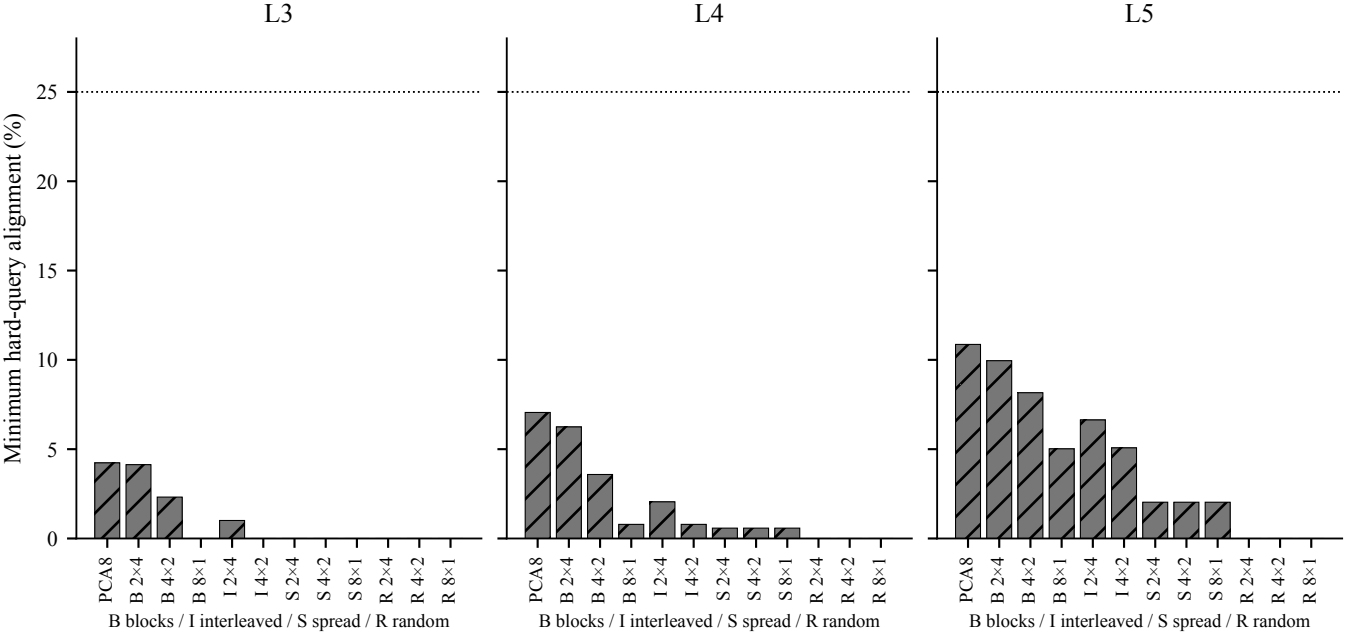

L3: 1,000 objects/region
L4: 100 objects/region
L5: 10 objects/region
Hard-6 alignment (%)
60
50
40
30
20
10
0
PCA blocks
Interleaved
Spread32
Random
55% ancestor gate
1×8
2×4
4×2
8×1
Views × dimensions (64 B)


Different directions do not rescue the hard-query canary
Query ID (* = hard)
660835
90208
359924 *
374533 *
96985 *
702545 *
55715 *
470632 *
94.8
6.9
93.3
29.9
90.3
5.2
0.0
0.0
77.9
0.0
1.2
53.4
0.1
0.0
0.0
0.0
6.2
0.0
2.0
0.0
0.7
0.0
0.0
0.0
22.0
0.1
19.9
0.1
10.5
0.1
0.0
0.0
31.9
0.1
28.4
0.1
10.4
0.1
0.0
0.0
16.4
0.0
4.7
2.8
0.6
0.0
0.0
0.0
35.4
0.2
23.1
4.3
20.4
0.2
0.0
0.0
49.1
4.7
56.5
0.1
40.2
4.7
0.0
0.0
PCA blocks V1
PCA blocks V2
Interleaved V1
Interleaved V2
Spread32 V1
Spread32 V2
Random V1
Random V2
Frozen 2×4 families at L4
Constituent pruned / truth-prunable (%)
100
80
60
40
20
0


Most pruning comes from one view; overlap is not new coverage
Pruned object-query pairs / hard-6 full scan (%)
35
30
25
20
15
10
5
0
Only this view
Also another view
PCA blocks V1
PCA blocks V2
Interleaved V1
Interleaved V2
Spread32 V1
Spread32 V2
Random V1
Random V2
Frozen 2×4 families at L4


L3
L4
L5
Minimum hard-query alignment (%)
0
5
10
15
20
25
PCA8
B 2×4
B 4×2
B 8×1
I 2×4
I 4×2
S 2×4
S 4×2
S 8×1
R 2×4
R 4×2
R 8×1
B blocks / I interleaved / S spread / R random